Analysis

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June 24, 2024

0.1 Analysis

Here there will be some theorems from Analysis I,II,III by Amann, Escher.
E.g. necessary def for "f(x+0)" is limit with certain base.
I prefer topological definition of limit and bases.
There should be some kind of theorem that will says that limit of all sequences in some class of sequences(depending on the base) is equal to the limit of function w.r.t. the base.
Potential theorem: in (maybe even pseudo?)metric spaces, "sequential limit"(limit of a sequence) is equal to the limit w.r.t some base.
There are wildly used esilon-delta-definitions of these limits. It would be nice to list theorems about that they are equivalent to limits w.r.t. bases.

Theorem 0.1.

$$(\forall \varepsilon > 0 : a + \varepsilon \geqslant 0) \to (a \geqslant 0)$$

Proof.

$$a < 0 \implies \exists \varepsilon > 0 : a + \varepsilon < 0$$
$$\varepsilon := \left(-\frac{a}{2}\right)$$
$$a < 0 \implies \varepsilon > 0$$
$$a + \varepsilon = \left(\frac{a}{2}\right) = -\varepsilon < 0$$

Done.